Class Imbalance in Histopathology: What Separates BRACS from TCGA-UT Before Any Imbalance

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Abstract

With balanced training data, a logistic-regression classifier on frozen Virchow2 features reaches 72.1% patient-macro balanced accuracy on TCGA-UT but only 39.6% on BRACS. This report explains what the two tasks ask, what a single image patch contains, and what the frozen representation and the fitted classifier distinguish. In the original feature space, the share of a test patch's 15 nearest training patches that carry its own class, adjusted for chance, is 0.55 on TCGA-UT and 0.20 on BRACS. The difference persists for cosine and Euclidean distance, for 5 to 30 neighbours, and with the same training support on both datasets. Classes with higher neighbour agreement are also recognised more reliably by the classifier. On BRACS, 74% of the patch variation within a class lies within individual patients, against 52% on TCGA-UT. The classifier's preference for the true class over its strongest rival is measured as a log-probability difference in nats; one nat corresponds to a probability ratio of $e \approx 2.7$. On BRACS, 67% of predictions, weighted so that each patient and class counts equally, lie within one nat of a tie, against 25% on TCGA-UT. When the most frequent class receives 100 times as many training patches as the rarest, BRACS loses 46% of the predictions the balanced classifier got right, and TCGA-UT loses 10%. Both fix a similar share of their balanced errors, 17% and 15%. On both datasets the losses concentrate on near-tie predictions, but at equal margin BRACS still loses more. Taken together, the accuracy gap reflects how well a single patch represents its label. A TCGA-UT patch shows tissue typical of its cancer type, and confusion stays within a few related classes. Many BRACS tiles show tissue that several lesion classes share, so their features mix across classes and the classifier decides close to a tie. Field of view, label inheritance from heterogeneous regions and class taxonomy all differ between the datasets and are not separated here.

1 Introduction

Both datasets pass through the same pipeline. Each image patch is encoded once by the frozen Virchow2 foundation model (Zimmermann et al., 2024), and a multinomial logistic regression is fit on the resulting 2,560-dimensional features. The headline metric is *patient-macro balanced accuracy*. A patient's recall in a class is the share of their patches in that class that the classifier labels correctly. These patient recalls are averaged within each class, so every patient counts equally regardless of how many patches they contribute. The resulting class recalls are then averaged with equal weight. A patient whose tissue carries two lesion classes contributes once to each class. With balanced training data this metric reaches 72.06% on TCGA-UT and 39.57% on BRACS, averaged over three patient-disjoint splits with ten training draws each (Queisler, 2026i,h). The chance level, one over the number of classes, is 3.3% for TCGA-UT's 30 classes and 14.3% for BRACS's seven.

These two numbers describe two experimental regimes, not an intrinsic ranking of the datasets. TCGA-UT trains on 20 patients and 640 patches per class, BRACS on 10 patients

and 320 patches. One BRACS input spans about 64 µm of tissue, one TCGA-UT input between 128 and 256 µm. Each test split contains 1,067 TCGA-UT patients against 24 BRACS patients.

Earlier studies on the same design show that class imbalance damages BRACS more than TCGA-UT (Queisler, 2026i,h). When a class becomes rare, two things change. Its examples contribute less to the training loss, so frequent classes shape the decision boundary more; this is the *frequency* effect. The classifier also sees fewer distinct examples of the rare class; this is the *support* effect. Both effects are larger on BRACS (Queisler, 2026b,a). Attempts to explain the difference stopped at their pre-registered gates for different reasons. Shifting the balanced classifier's stored decisions by the imbalanced class shares overstated the damage of refitted classifiers (Queisler, 2026g). Increasing BRACS's class separation increased rather than reduced its damage (Queisler, 2026j). BRACS's classes fell outside TCGA-UT's range of class properties, so class composition could not be separated from dataset identity (Queisler, 2026c). A joint separation and centre-correction rescue did not survive fixed regularization (Queisler, 2026e). Patch reselection could not change support coverage enough to test its contribution (Queisler, 2026k). Better regularization selection closed less than half of the gap in frequency-only damage (Queisler, 2026f). With frozen UNI2-h features instead of Virchow2, the damage changed by less than one point (Queisler, 2026d). None of these outcomes refutes the proposed explanations; each leaves its explanation untested or only partly supported. This report returns to the balanced condition. It asks what differs visibly and measurably before imbalance, then examines how those differences appear under imbalance.

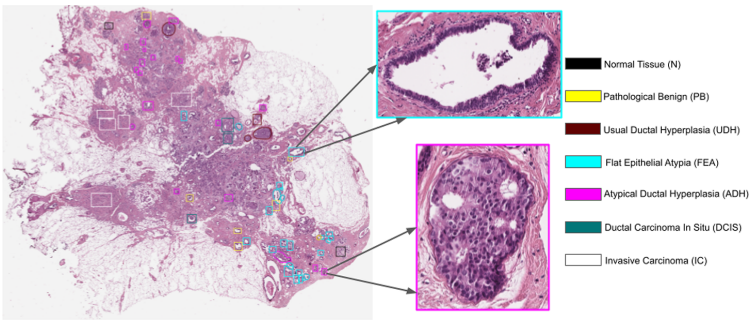
2 Background

Haematoxylin and eosin (H&E) staining colours cell nuclei purple-blue and most other tissue pink. In both datasets the relevant structures are nuclei, sheets of epithelial cells that line glands and ducts, and the surrounding connective tissue, or stroma. In the breast, epithelium lines a branching system of ducts and lobules embedded in stroma and fat.

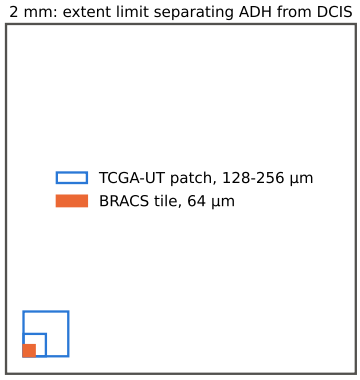
TCGA-UT asks *which cancer is this*, across 30 cancer types from many organs. Its patches were cropped by the dataset authors from tumour regions that pathologists had delineated as uniform, so most patches show cancerous tissue typical of the source slide (Komura et al., 2022). Many classes differ by organ, and organs differ in tissue architecture; some do not. Colon and rectal adenocarcinoma, for example, arise from the same type of intestinal lining.

BRACS asks *which kind of breast lesion is this*, among seven categories: normal tissue (N), pathological benign lesions (PB), usual ductal hyperplasia (UDH), flat epithelial atypia (FEA), atypical ductal hyperplasia (ADH), ductal carcinoma in situ (DCIS) and invasive carcinoma (IC) (Brancati et al., 2022). These categories are clinical groupings, not points on one linear disease scale: PB alone combines more than ten distinct benign entities, and UDH, ADH and DCIS all grow inside ducts. The dataset paper names one pair as especially hard. ADH has cytological atypia similar to that of low-grade DCIS but spans no more than 2 mm or only partially involves ducts; in some cases it "includes all the features of DCIS, but is simply limited in size" (Brancati et al., 2022). During annotation, pathologists sometimes had to examine the whole slide to decide between such classes.

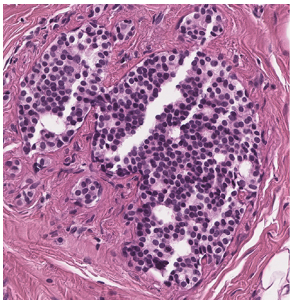
Figure 1 shows why the size of an input matters. A BRACS region of interest is cut into non-overlapping 256×256-pixel tiles at the scanner's native 0.25 µm per pixel, and every tile inherits the region's label. Each tile therefore covers 64×64 µm, which is a few cell diameters. Virchow2's preprocessing resizes the whole tile to 224×224 pixels without cropping. A region of ADH can be up to 2 mm across, roughly thirty tiles in each direction, and some of its tiles show only stroma between the atypical ducts. The 2 mm extent criterion that separates ADH from DCIS cannot be seen inside any single tile. TCGA-UT patches cover two to four times more tissue in each direction, and they were selected from uniform tumour areas (Komura et al., 2022). These differences are candidate contributors to the accuracy gap.



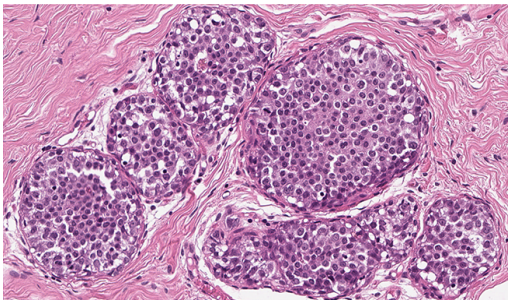
(a) Slide with annotated regions of interest



(b) Field of view of one model input



(c) ADH region



(d) DCIS region

Figure 1: A BRACS label belongs to a whole annotated region, while the model sees one small tile. (a) A BRACS slide with its annotated regions; colour marks the lesion class. (b) Physical extent of one model input against the 2 mm extent that separates atypical ductal hyperplasia (ADH) from ductal carcinoma in situ (DCIS), drawn to scale: a BRACS tile spans 64 µm, a TCGA-UT patch 128–256 µm. (c, d) Published example regions of ADH and DCIS. The dataset paper describes ADH as monomorphic cells that only partially fill the ductal spaces, and DCIS as a proliferation that fills the entire duct. Panels (a), (c) and (d) are reproduced from Brancati et al. (2022), Figures 1 and 3(e, f), under CC BY-NC 4.0 (`https://creativecommons.org/licenses/by-nc/4.0/`); (c) and (d) are cropped, and the source gives no scale. Panel (b) is a schematic, not an image.

3 Methods

The analyses use the test predictions of classifiers trained under four conditions in earlier studies (Queisler, 2026i,h,b,a). In the *balanced* condition, every class has the same number of training patients and patches. The other three conditions set the ratio ρ between the most and the least frequent class to 100 at the same total patch budget. The *ratio* condition (r100) draws imbalanced training data and thus combines the frequency and support effects. The *frequency-only* condition (F100) keeps the balanced training patches but weights each class by its imbalanced share. The *support-only* condition (S100) keeps the imbalanced patches but weights all classes equally. Each condition has three patient-disjoint splits and ten training draws per split, giving 30 classifiers per condition and dataset; each receives equal weight. Recomputed balanced accuracies reproduce the published values. Results are descriptive point estimates; the analysis computes no new intervals.

Confusion and margins. Confusion matrices use the same patient weighting as the headline metric, so each row sums to one and the mean diagonal equals balanced accuracy. The *margin* of a test patch is $\log p_{\text{true}} - \max_{d \neq \text{true}} \log p_d$, the classifier's log-probability preference for the true

class over its strongest rival. A positive margin means a correct prediction. Margins use natural logarithms and are given in nats: a margin of one nat means that the true class is $e \approx 2.7$ times as probable as its strongest rival, and a margin of zero is a tie. Margins are also shown after dividing by each classifier's validation temperature (Guo et al., 2017).

Neighbour agreement. For each split, the analysis draws a reference set from training patients: 10 patients per class with 32 randomly chosen patches each, so every class has equal mass. It draws a query set from the same split's test patients: up to 10 patients per class and up to 32 patches each, weighted per patient and class. The reference therefore has the same support on both datasets, 320 patches from 10 patients per class. For each query patch, *neighbour agreement* is the share of its k nearest reference patches that carry its own class. The analysis uses the full 2,560-dimensional features, measuring cosine distance on normalised features and, separately, Euclidean distance on raw features. The main setting is $k = 15$, with $k = 5$ and $k = 30$ as sensitivity checks. Because the datasets have different class counts K, the analysis also reports chance-adjusted agreement, $(a - 1/K)/(1 - 1/K)$, where a is raw neighbour agreement. This quantity is 0 at chance and 1 for perfect agreement. For each class in the same reference sets, the analysis also computes the share of within-class patch variance that lies within individual patients. Principal component analysis (PCA), fitted on the reference set and applied to the query set, illustrates the structure in two dimensions.

Examples. A fixed rule selects example tiles from the first split. For each class, it selects the query patch whose neighbour agreement is closest to the class median, together with its nearest reference patch from the confusion partner (ADH and DCIS, colon and rectal adenocarcinoma) and from any other class. The pairs follow the BRACS dataset paper and an earlier TCGA-UT study that identified colon and rectal adenocarcinoma as confusion partners (Brancati et al., 2022; Queisler, 2026l). The best-recognised class serves as a counterexample. The tiles are copied unaltered from the image archives that produced the features.

4 Representation geometry

TCGA-UT test patches sit among training patches of their own class far more often than BRACS patches do. At $k = 15$ with cosine distance, raw agreement is 0.56 on TCGA-UT against a chance level of 0.03, and 0.32 on BRACS against 0.14. After chance adjustment this is 0.55 against 0.20 (Figure 2). The difference is stable across all three splits (TCGA-UT 0.54–0.56, BRACS 0.20–0.22), across k from 5 to 30, and across both distances: Euclidean distance on the raw features gives 0.55 and 0.20. Because both reference sets have the same support, the difference does not stem from BRACS's smaller training cohort. Within TCGA-UT, classes range from 0.13 to 0.97; all seven BRACS classes fall below 0.45.

Classifier performance follows this geometry. Across classes, neighbour agreement and balanced-condition recall have a Spearman rank correlation of 0.96 on TCGA-UT (30 classes) and 0.82 on BRACS (7 classes). Neighbour agreement uses a fixed training sample and no classifier, so the two measures do not derive from the same predictions. Classes with mixed neighbourhoods are also harder for the classifier to recognise.

Two further measurements describe how the mixing is organised. First, on BRACS 74% of the within-class patch variance lies within individual patients, against 52% on TCGA-UT. A BRACS patient's own tiles are thus nearly as varied as tiles from different patients, consistent with every tile inheriting the label of a heterogeneous region. Second, an earlier study reported a class-centre separation index of 0.44–0.48 on BRACS and 0.68–0.71 on TCGA-UT (Queisler, 2026j). The index divides the median distance between nearest class centres by the pooled spread of *patient means*. It measures how far apart class averages lie relative to patient-to-

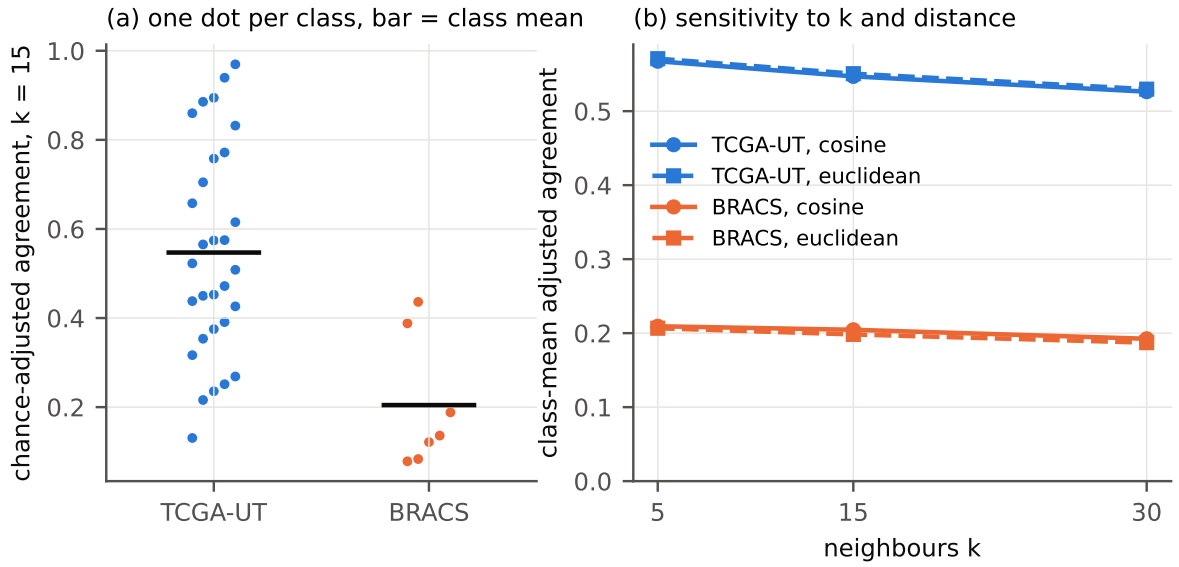


Figure 2: In the original feature space, BRACS test patches have far fewer same-class neighbours among training patches than TCGA-UT patches do. (a) Chance-adjusted share of the 15 nearest training patches with the query's class, one dot per class, averaged over three splits; bars mark class means. (b) Class means for $k = 5$, 15 and 30, for cosine (circles, solid) and raw Euclidean distance (squares, dashed); the two distances nearly coincide.

patient variation. Neighbour agreement measures how individual patches mix. Both indicate the same pattern but measure different quantities.

Figure 3 shows both confusion pairs in two dimensions. Colon and rectal adenocarcinoma occupy the same corner of TCGA-UT's projection, apart from most other classes. ADH and DCIS spread across BRACS's projection among all other classes.

5 Balanced decisions

On BRACS, only IC (69% recall) and normal tissue (60%) are recognised in most cases. ADH is recognised in 12% of the patient-macro mass, below the seven-class chance level. It is predicted as FEA (23%), UDH (21%) and DCIS (18%) more often than as itself, and DCIS in turn is predicted as ADH in 18% (Figure 4a). Normal tissue and PB exchange 23% and 32% of their mass. On TCGA-UT, recall ranges from 28% for oesophageal carcinoma to 95% for thyroid carcinoma, and 21 of 30 classes exceed 65% (full matrix in Appendix A). The classifier predicts colon adenocarcinoma as rectal adenocarcinoma in 36% of the patient-macro mass, and rectal as colon in 27%. It predicts oesophageal carcinoma as head-and-neck squamous cell carcinoma in 19% and as stomach adenocarcinoma in 15%. On average, a class's strongest rival takes 34% of its errors on TCGA-UT and 35% on BRACS. With 29 possible rivals on TCGA-UT, that is ten times the share an even spread would give; with six on BRACS, it is twice. TCGA-UT's errors thus concentrate on a few anatomically or histologically related classes, while BRACS's spread over most of the matrix.

The margin distribution shows how decisive the classifier is (Figure 4b). On TCGA-UT the median of the patient-macro margin distribution is 1.65 nats: the typical patient's true class is about five times more probable than its strongest rival. On BRACS the median is -0.20 nats, so for more than half of the patient-macro mass a rival class wins. Two thirds of BRACS's mass lies within one nat of a tie, against a quarter of TCGA-UT's. Temperature scaling stretches both distributions on average, because the mean fitted temperatures are below one (0.65 and 0.84;

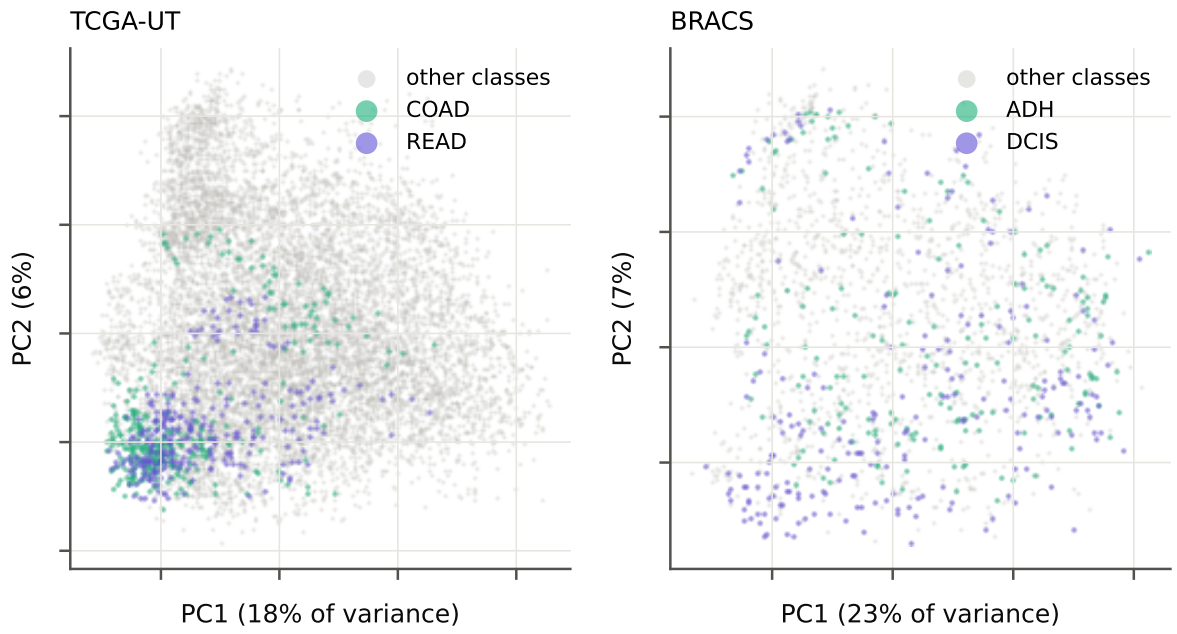
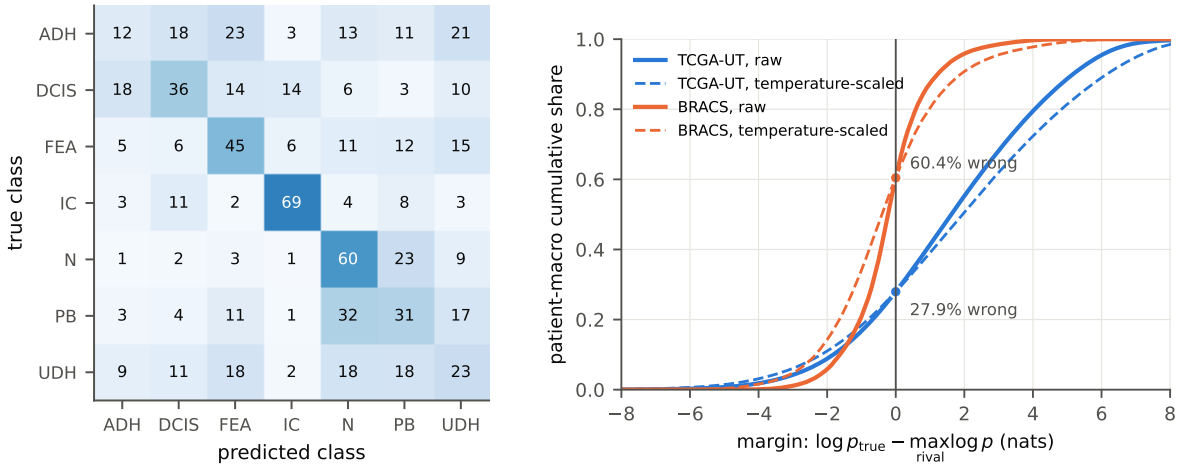


Figure 3: A two-dimensional projection shows the confusion pairs overlapping on both datasets. Held-out test patches from the first split are projected onto each dataset's own principal component analysis (PCA) axes, fitted on that split's balanced training sample. The highlighted classes form each dataset's confusion pair; all other classes are grey. The two panels have separate axes, so distances and cluster areas cannot be compared between panels. The first two components explain 23% (TCGA-UT) and 30% (BRACS) of the variance.



(a) BRACS confusion (%)

(b) Margin distribution

Figure 4: BRACS errors are spread over most class pairs, and most BRACS predictions are close to a tie. (a) Patient-macro confusion of the balanced classifiers on BRACS; each row sums to 100% and the diagonal is per-class recall. (b) Patient-macro cumulative distribution of the true-class margin, raw (solid) and after validation temperature scaling (dashed). The value at zero is the error rate, one minus balanced accuracy.

individual classifiers range from 0.30 to 1.19), but it does not change their ordering or the sign of any margin. Weighting matters here: pooling patches without patient weighting puts BRACS's median at +0.32 nats (averaged over classifiers), because each patient then counts in proportion

to its number of patches. Patch-pooled margin summaries from earlier work (Queisler, 2026g) are therefore not patient-macro results.

Figure 5 makes these summaries concrete. The colon and rectal adenocarcinoma tiles show glandular tumour tissue of similar appearance, and each test tile's nearest partner-class tile lies at a cosine distance of only 0.09–0.11. The median-agreement ADH tile shows mainly pink stroma with few nuclei. It carries the ADH label because it was cut from an ADH region, and none of the ten balanced classifiers classifies it correctly. Its nearest DCIS training tile is also mostly stroma. The DCIS tile and its nearest ADH neighbour both show crowded, rounded nuclei, the shared cytology the dataset paper describes. These examples show that a 64 µm tile can lack the structure on which its inherited label depends.

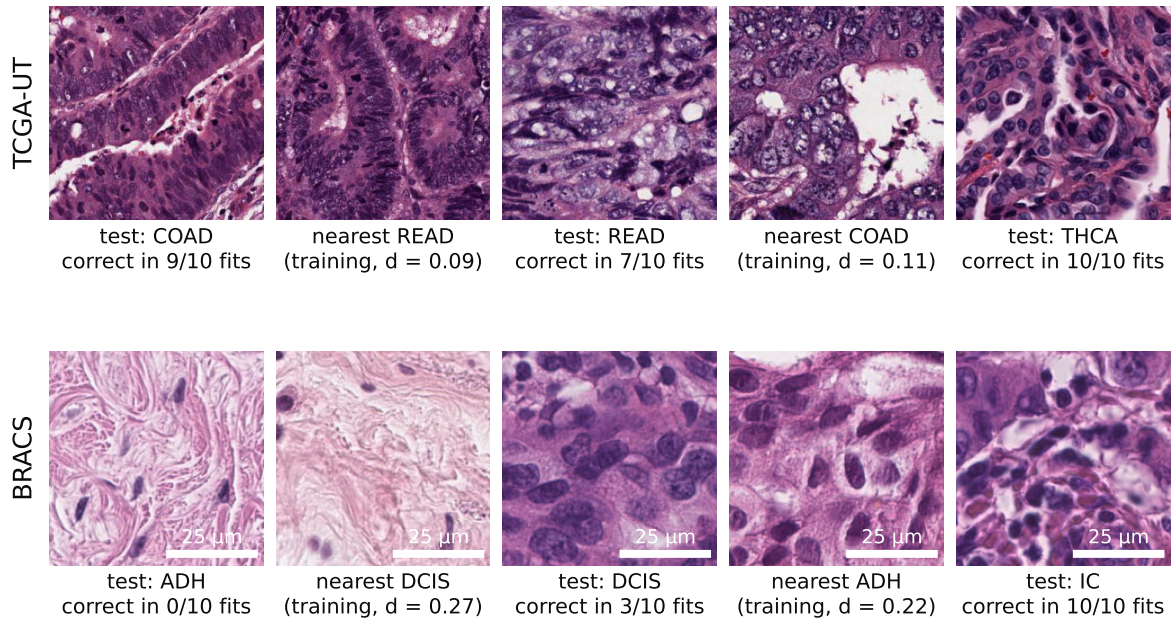


Figure 5: Example test tiles and their nearest training neighbours from the confusion partner. Each test tile is the one closest to its class's median neighbour agreement; "correct in $n/10$" counts the balanced classifiers of the first split that classify it correctly, and d is the cosine distance. The right column shows the best-recognised class. BRACS tiles carry a 25 µm scale bar; TCGA-UT patches span 128–256 µm, recorded per dataset rather than per patch, so they carry none.

6 Prediction changes under imbalance

At $\rho = 100$, total damage is 7.65 points on BRACS and 2.91 on TCGA-UT; the frequency-only and support-only conditions isolate each effect (Queisler, 2026b,a). Because balanced and imbalanced classifiers predict the same test patches, each change in balanced accuracy splits exactly into balanced-correct predictions that are lost and balanced errors that are fixed (Figure 6). The net damage is much smaller than the turnover on both datasets. In the ratio condition, BRACS loses 18.1 points of correct predictions and gains 10.4; TCGA-UT loses 7.2 and gains 4.3.

Relative losses of balanced-correct predictions differ more than relative gains from balanced errors. In the ratio condition, BRACS loses 46% of its balanced-correct mass and TCGA-UT 10%. Of their balanced errors, BRACS fixes 17% and TCGA-UT 15%. Table 1 splits the balanced-correct mass by its balanced margin. On both datasets the loss rate falls steeply with margin, from 62% (BRACS) and 45% (TCGA-UT) for margins up to 0.5 nats to 8% and 2%

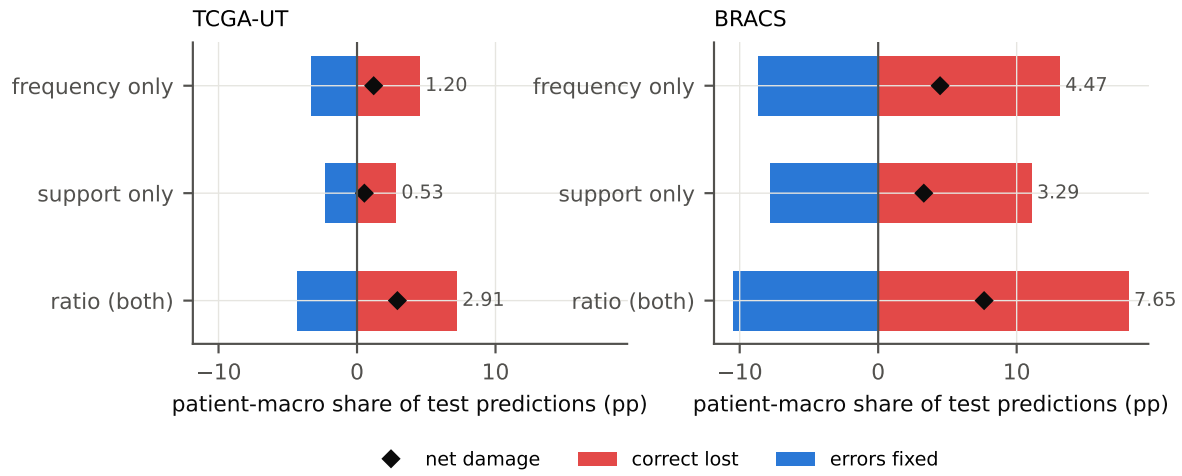


Figure 6: Imbalance moves many more BRACS predictions in both directions. For each condition at $\rho = 100$, balanced-correct predictions that become wrong (red, right) and balanced errors that become correct (blue, left), in patient-macro percentage points; the diamond and number give the net damage, which equals the difference exactly.

between 2 and 4 nats. Because 68% of BRACS's correct mass lies within one nat of a tie, against 19% of TCGA-UT's, that region supplies 85% of BRACS's lost mass and 70% of TCGA-UT's. In every margin bin, however, BRACS loses more than TCGA-UT, so the different margin distributions describe much, but not all, of the difference. The frequency-only and support-only conditions show the same pattern: 93% and 91% of BRACS's lost mass, and 86% and 89% of TCGA-UT's, come from within one nat.

	Correct mass (pp)		Lost at $\rho = 100$ (%)	
Balanced margin (nats)	TCGA-UT	BRACS	TCGA-UT	BRACS
0–0.5	6.5	17.4	45	62
0.5–1	6.9	9.6	29	48
1–2	13.9	8.5	13	29
2–4	24.2	3.8	2	8
> 4	20.5	0.3	0	0
Total	72.1	39.6	10	46

Table 1: Losses concentrate on near-tie predictions on both datasets, but BRACS loses more at every margin. Balanced-correct patient-macro mass in percentage points (pp) by its balanced margin, and the share of it lost in the ratio condition at $\rho = 100$, averaged over 30 classifiers per dataset.

7 Discussion

The balanced condition separates the two datasets consistently (Table 2). On BRACS, patches of different classes mix in the frozen feature space, and the patches of a single patient vary nearly as much as patches across patients. The balanced classifier spreads its errors across most class pairs, and most of its decisions lie near a tie. On TCGA-UT, confusion is concentrated in a few related pairs, and most decisions are clear. Under imbalance, losses concentrate on near-tie correct predictions on both datasets, although BRACS loses more than TCGA-UT even at equal margin.

Quantity	TCGA-UT	BRACS
Earlier studies		
Balanced accuracy, balanced condition (%)	72.06	39.57
Damage at $\rho = 100$, ratio condition (pp)	2.91	7.65
frequency only	1.20	4.47
support only	0.53	3.29
This report		
Chance-adjusted agreement with the 15 nearest training patches	0.55	0.20
Rank correlation of class agreement with class recall	0.96	0.82
Within-patient share of within-class variance (%)	52	74
Strongest rival's share of a class's errors (%)	34	35
Median margin (nats)	1.65	−0.20
Mass within one nat of a tie (%)	25	67
Balanced-correct predictions lost at $\rho = 100$ (%)	10	46
Balanced errors fixed at $\rho = 100$ (%)	15	17
Lost mass from within one nat of a tie (%)	70	85

Table 2: BRACS differs from TCGA-UT in most measures; only the strongest rival's share of errors and the share of errors fixed under imbalance are similar. Earlier results come from Queisler (2026i,h,b,a); all other values are descriptive point estimates, patient-macro where they refer to predictions.

These observations point to one reading of the accuracy gap: a BRACS tile often does not show what its label refers to. Each tile covers 64 µm and inherits the label of a region that may span millimetres, so many tiles show stroma or cytology that several lesion classes share. The frozen features then place such tiles among other classes, and the classifier can only decide close to a tie. TCGA-UT patches are larger and come from uniform tumour areas, and most of its classes differ in organ and tissue architecture. Three candidate causes fit this reading: the small field of view, label inheritance from heterogeneous regions, and the finer clinical taxonomy. The datasets differ in all three at once, so their contributions cannot be separated. The neighbour comparison matches training support, but the smaller BRACS training cohort remains a candidate for part of the accuracy gap itself.

The link to imbalance damage is less direct. Small balanced margins describe where most damage occurs, but they do not predict how much a refitted classifier loses. An earlier study showed that shifting the stored balanced probabilities by the imbalanced class shares overstates the damage of refitted classifiers by a factor of three to nine (Queisler, 2026g). Another study showed, in a pilot with one training draw per split, that artificially increasing BRACS's class separation raised its balanced validation accuracy but increased its total imbalance damage (Queisler, 2026j). Better balanced separation therefore does not imply smaller imbalance damage, and why BRACS loses more even at equal margin remains open.

8 Limitations

The datasets are not matched in training support, field of view, class count or taxonomy. Chance adjustment removes only the trivial effect of the class count; it does not make the two taxonomies comparable. All values are descriptive point estimates without intervals, and patches are never treated as independent samples. Margins are score-based summaries whose scale depends on regularization and calibration; a margin is not a distance in feature space.

The neighbour analysis uses at most 10 patients per class and split. On BRACS this is the whole feasible test cohort for several classes, so per-class values rest on few patients. The reference includes only patients with at least 32 patches in a class, which favours patients with large annotated regions. Numerical statements rest on the full feature space only; the PCA

panels have separate axes and retain a minority of the variance, so they illustrate rather than measure. The example tiles are post hoc selections that illustrate the summaries but do not estimate how often each pattern occurs. Their visual descriptions follow the dataset paper and were not confirmed by a pathologist.

All results are conditional on frozen Virchow2 features and a logistic-regression classifier. Separating the candidate causes would require experiments that vary one factor at a time, such as the BRACS field of view or the number of training patients.

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A TCGA-UT confusion matrix

Figure 7 gives the full patient-macro confusion matrix of the balanced TCGA-UT classifiers. The largest off-diagonal cells link related classes: colon and rectal adenocarcinoma, lower-grade glioma (LGG) and glioblastoma (GBM), cutaneous melanoma (SKCM) and uveal melanoma (UVM), oesophageal, stomach and head-and-neck carcinoma, and the squamous-cell carcinomas of head and neck, lung and cervix.

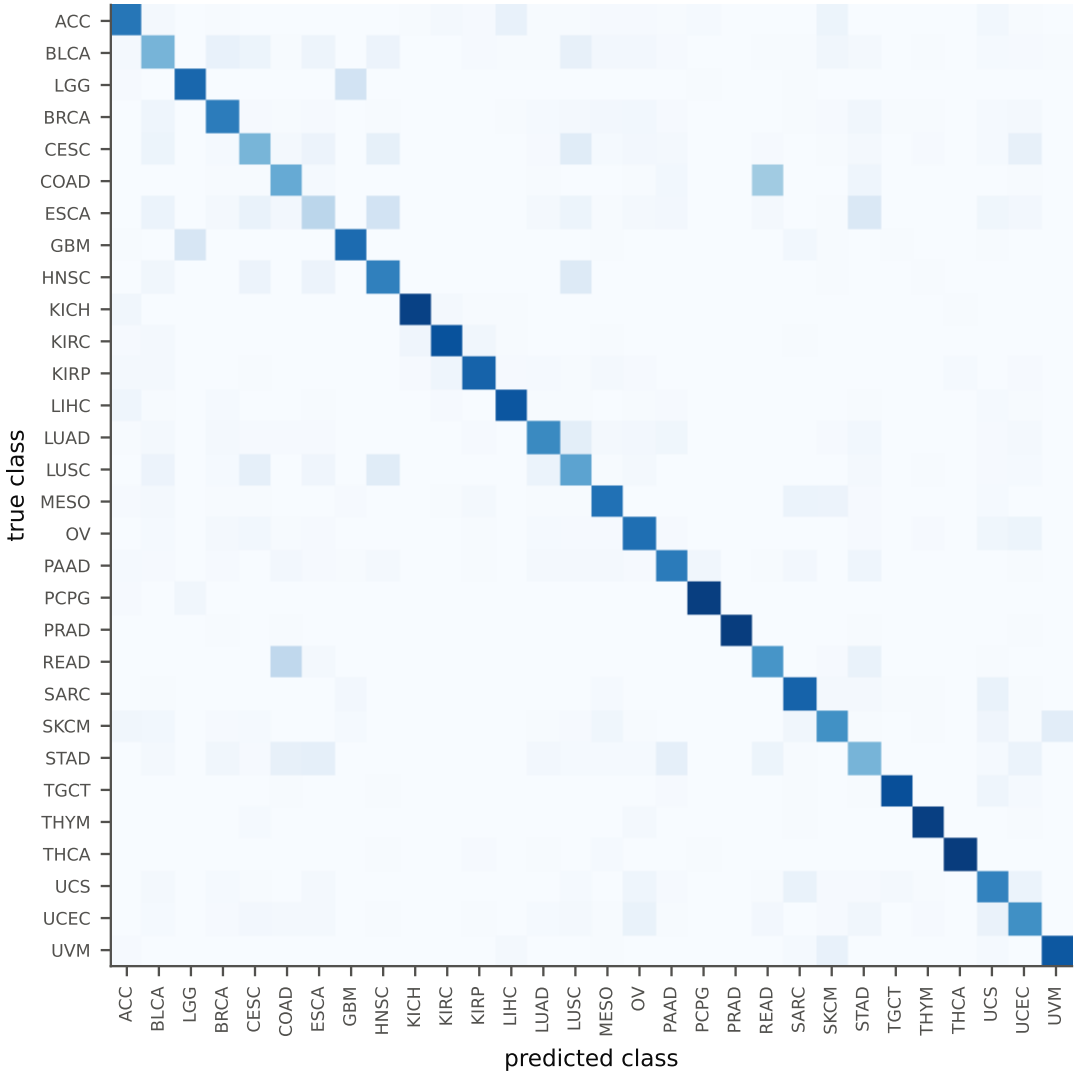


Figure 7: Patient-macro confusion of the balanced classifiers on TCGA-UT (rows: true class, summing to 100%; TCGA study abbreviations). Darker cells carry more mass.